

Fourfold forrelation at $N = 4096$: a construction and a conditional lower bound

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Abstract

Three single photons, each crossing two phase masks, solve the fourfold-forrelation promise with hard dose six and error $81/256$. We reduce a proposed parallel lower bound to explicit occurrence-kernel inequalities and a finite spectral calculation. The spectral arithmetic is reproducible, but supplied coefficient derivations contain errors: the finite lower bound remains conditional. We also correct its probe restriction from physical photon number to parity-support size.

1 The task and the status of the results

For $N = 2^d$, index coordinates by $\mathbb{F}_2^d$ and set $H_{ij} = N^{-1/2}(-1)^{i \cdot j}$, $u = N^{-1/2}(1, \dots, 1)^\top$. Four binary phase masks $D_b = \text{diag}(x^{(b)})$ define

$$F(x) = \text{Forr}_4(x) = u^\top D_1 H D_2 H D_3 H D_4 u. \quad (1)$$

Decide $F \geq 1/4$ versus $F \leq -1/4$ with worst-case error at most $1/3$. This is a variant of the query problem of Aaronson and Ambainis.¹ Dose counts every photon-mask traversal and bounds the entire support of the signal state, not its mean or a postselected cost.

The two-pass upper bound below is unconditional. The proposed finite obstruction concerns one parallel interrogation: a fixed probe passes once through the direct-sum mask and is measured by an arbitrary POVM. References and idlers bypass the masks. For a Fock occupation ν , define

$$S_b(\nu) = \{i : \nu_{b,i} \text{ is odd}\}, \quad Q|\nu\rangle = \left(\sum_b |S_b(\nu)| \right) |\nu\rangle.$$

The balanced coefficient ledger requires

$$[\rho, Q] = 0, \quad \rho \text{ supported on } \hat{N}_{\text{sig}} \leq 6. \quad (2)$$

Thus the probe is block diagonal in parity-support size. Repeated modes and idlers are allowed. Fixed physical photon number alone is insufficient: the superposition of three photons in one mode of block one and one photon in each of blocks two, three and four has parity sizes one and three. The components acquire different phase characters.

Theorem 1 (Conditional finite obstruction). *Assume the complete-kernel bounds (11) hold with the recorded coefficients. At $N = 4096$ there are laws P_+, P_- on the respective promises such that every parallel probe satisfying (2) has output laws obeying*

$$\text{TV}(T_+, T_-) \leq 0.260969224792207925. \quad (3)$$

Its equal-prior Bayes error is at least $0.369515387603896037 > 1/3$.

¹S. Aaronson and A. Ambainis, *Forrelation: a problem that optimally separates quantum from classical computing*, STOC 2015, pp. 307–316.

The hypothesis is not established: supplied local-Walsh and four-cubic intermediate bounds are false. See `../AUDIT.md` for counterexamples.

2 The two-pass protocol

Prepare $(|0\rangle + |1\rangle)u/\sqrt{2}$. In arm zero apply D_1, H, D_2 ; in arm one apply D_4, H, D_3, H . The resulting state is $(|0\rangle L_x + |1\rangle R_x)/\sqrt{2}$, where

$$L_x = D_2 H D_1 u, \quad R_x = H D_3 H D_4 u.$$

Each arm crosses exactly two masks. Recombining the paths gives

$$\Pr(X = \pm 1 \mid x) = \frac{1 \pm \langle L_x, R_x \rangle}{2} = \frac{1 \pm F(x)}{2}. \quad (4)$$

Discard the mode label and retain the output port. On the promise, each flag is correct with probability at least $5/8$. Three independent flags and majority decoding have error at most

$$(3/8)^3 + 3(5/8)(3/8)^2 = 81/256 < 1/3,$$

and deterministic dose $3 \cdot 2 = 6$. Equality holds at either promise boundary. No outcome is discarded. This construction works for every N .

3 The hard inputs

Set $q = 64$, so $N = q^2$. Identify a coordinate with $(a, b) \in [q]^2$, and use row-major flattening, so $H_N = H_q \otimes H_q$. Write $S_q = \sqrt{q} H_q$. For a uniform permutation π and independent uniform signs σ_b , define arrays

$$L_{a,b} = S_q(a, \pi(b))\sigma_b, \quad R_{\pi(b),a} = \sigma_b S_q(b, a). \quad (5)$$

Their useful property is simply

$$H_N L = R, \quad H_N R = L. \quad (6)$$

Indeed, at coordinate (c, d) ,

$$(H_N L)_{c,d} = \frac{1}{q} \sum_{a,b} S_q(c, a) S_q(d, b) S_q(a, \pi(b)) \sigma_b = S_q(d, b_c) \sigma_{b_c} = R_{c,d},$$

where $b_c = \pi^{-1}(c)$. The second identity follows from $H_N^2 = I$.

Take three independent pairs (L_j, R_j) and set

$$z = (L_1, R_1 \odot L_2, R_2 \odot L_3, R_3). \quad (7)$$

Starting on the right of (1), (6) successively cancels $R_3, L_3, R_2, L_2, R_1, L_1$ and leaves u . Thus $F(z) = 1$ exactly. Negating the first block changes the value to -1 .

Now independently flip every coordinate with probability $3/25$. Equivalently, multiply it by a sign η with mean

$$\beta = 19/25.$$

Let $\tilde{P}_+$ be the resulting positive law and $\tilde{P}_-$ its first-block reflection. Then $\mathbb{E}_{\tilde{P}_+} F = \beta^4$. Finally condition each law onto its own promise to obtain P_+, P_- .

Lemma 2 (Cost of conditioning). *The conditioning loss is at most 0.002225128406630703.*

Proof. Fix an exact plant z . Reveal the four noisy blocks in order from left to right and form the Doob martingale of F . When block b is exposed, the future $4 - b$ masks contribute their means, giving a factor β^{4-b} . By (6), the remaining deterministic right-hand chain is a sign vector divided by $\sqrt{N}$. The left-hand chain has norm one. The sum of squared coefficients of the newly exposed signs is therefore $\beta^{2(4-b)}/N$.

A sign in $[-1, 1]$ has centered subgaussian proxy at most one by Hoeffding's lemma. Conditional independence within each newly exposed block gives the martingale proxy

$$V_\beta = \frac{1 + \beta^2 + \beta^4 + \beta^6}{N} = \frac{(1 + \beta^2)(1 + \beta^4)}{N}.$$

With $g = \beta^4 - 1/4$ and $x = g^2/(2V_\beta)$, Chernoff gives a failure probability at most e^{-x} . This holds for every plant. Reflection gives the same bound on the negative side. Conditioning changes each input law in total variation by at most e^{-x} . Data processing and the triangle inequality therefore add at most $2e^{-x}$ to the output total variation. The verifier evaluates the exact rational upper

$$2e^{-x} \leq 2(1 + x/4096)^{-4096} < 0.002225128406630703.$$

□

4 From parity supports to one matrix

The mask acts on an occupation only through its parity support: $U_x|\nu\rangle = \chi_{S(\nu)}(x)|\nu\rangle$, where $\chi_S(x) = \prod_b \prod_{i \in S_b} x_i^{(b)}$. By (2), decompose the input into fixed- Q sectors and purify each sector separately. Group all Fock amplitudes with support S into a vector ψ_S , including idlers. These vectors are mutually orthogonal; put $p_S = \|\psi_S\|^2$. For the unconditioned hard laws define

$$K_{ST} = \frac{1}{2}(\mathbb{E}_{\tilde{P}_+} - \mathbb{E}_{\tilde{P}_-})\chi_{S \Delta T}. \quad (8)$$

The map $|S\rangle \mapsto \psi_S/\sqrt{p_S}$ is an isometry on nonzero weights. Hence half the difference of the averaged purified output states has trace norm exactly

$$\|\text{diag}(\sqrt{p})K\text{diag}(\sqrt{p})\|_1. \quad (9)$$

Discarding a purifier or measuring cannot increase this norm. Convexity handles sector mixtures. This reduction includes all repeated occupations, without identifying photon number with parity size.

The independent permutation signs force adjacent block degrees in (8) to have the same parity; first-block reflection selects the odd case. Thus every surviving symmetric-difference profile has odd entries. Define, block by block,

$$n_b = |S_b|, \quad m_b = |T_b|, \quad a_b = |S_b \Delta T_b|, \quad t_b = |S_b \setminus T_b|.$$

These are counts of parity-support labels. Since $|S| = |T|$ in one sector,

$$m = n + a - 2t, \quad 2|t| = |a|. \quad (10)$$

Here $4 \leq |a| \leq 12$, and independent noise contributes $\beta^{|a|}$.

5 The coefficient hypothesis and its spectral consequence

For an odd profile a and balanced split t , let the row labels A_b and column labels B_b be sets of sizes t_b and $a_b - t_b$. For the exact positive plant z in (7), define the *complete* marked kernel

$$\mathcal{O}_{a,t}(A, B) = \mathbf{1}_{\{A_b \cap B_b = \emptyset \ \forall b\}} \mathbb{E}_z \chi_{A \cup B}(z).$$

In particular, every distinctness constraint across the split is part of the matrix. The required analytic hypothesis is

$$\sup_{p,w} \|\text{diag}(\sqrt{p}) \mathcal{O}_{a,t} \text{diag}(\sqrt{w})\|_1 \leq \hat{c}_{a,t}, \quad (11)$$

for arbitrary normalized row and column laws p, w , with the rational coefficients used in the stored ledger.

Here is the full lifting argument. Fix n, m, a, t and write $u = a - t$. For each residual support R of block counts $n - t = m - u$, restrict A, B to avoid R and embed them as $S = R \cup A$, $T = R \cup B$. The sum of these embedded matrices is the desired occurrence matrix: a nonzero entry has the unique residual $R = S \cap T$. For normalized laws p_S, w_T , put $P_R = \sum_A p_{R \cup A}$ and $W_R = \sum_B w_{R \cup B}$. The contribution of each residual has weighted trace norm at most $\hat{c}_{a,t} \sqrt{P_R W_R}$ by (11). Triangle inequality and Cauchy–Schwarz give

$$\hat{c}_{a,t} \sum_R \sqrt{P_R W_R} \leq \hat{c}_{a,t} \sqrt{b(n, t) b(m, u)}, \quad b(n, t) = \prod_{j=1}^4 \binom{n_j}{t_j}.$$

Indeed each S appears in exactly $b(n, t)$ residuals, and each T in $b(m, u)$ residuals. This proves the combinatorial factor without omitting shared labels or distinctness masks.

Index a matrix A by $\mathcal{N} = \{n \in \mathbb{Z}_{\geq 0}^4 : |n| \leq 6\}$, which has $\binom{10}{4} = 210$ states. For every compatible profile–split pair (a, t) with $2|t| = |a|$, add

$$\beta^{|a|} \hat{c}_{a,t} \sqrt{b(n, t) b(m, a - t)} \quad (12)$$

to A_{nm} and set $B = (A + A^\top)/2$. Restoring the probability of each block-count sector, the preceding bound gives $(9) \leq v^\top B v$, where $v_n = \sqrt{\text{Pr}(n)}$ and $\|v\|_2 = 1$. It is therefore at most $\lambda_{\max}(B)$. The matrix is nonnegative and block diagonal in total parity-support size.

The ledger reconstructs B with directed decimal arithmetic and outward square roots. A stored positive vector w provides the bound

$$\lambda_{\max}(B) \leq \max_i \frac{(\hat{B}w)_i}{w_i} < 0.258744096385577223.$$

A floating eigensolve supplies only the candidate; the directed Collatz ratios supply the spectral certificate for the recorded matrix. Adding Lemma 2 gives (3) and the Bayes identity $(1 - \text{TV})/2$ proves the conditional theorem. The total is rounded after adding the high-precision components; adding their displayed upper bounds would be slightly looser.

The coefficient guide `code/COEFFICIENTS.md` maps the 888 high-sector rows and inherited inputs to their derivations and rounding rules. Rounding a source value upward does not prove (11); false or missing intermediate bounds leave that hypothesis unresolved. The broader finite and adaptive claims remain open. The root `README.md` gives the audit, build and replay instructions.